

**JENKINS IS BROKEN.
WHERE DO I START?**

WHERE TO LOOK

ACCESS

How users and agents reach Jenkins
Ingress, Service

`kubectl get ingress,svc,endpoints`

CONTROLLER

UI, jobs, queue, plugins
StatefulSet pod, init containers

`kubectl describe pod <pod>`
`kubectl logs <pod> -c <container>`

JENKINS_HOME

Config, builds, logs, artifacts
PersistentVolume

`kubectl get pvc`

AGENTS

Where builds run
Ephemeral pods

`kubectl describe pod <agent>`

CLUSTER

What everything runs on
Nodes, scheduler, storage, network

`kubectl get events -A`
`kubectl top nodes`

NO LEFT SPACE

BUILD NOW DOES NOTHING

Full before the build

NO BUILD.
NO CONSOLE.
NO AGENT POD.

Jenkins can't create the build directory, so the build never exists. Kubernetes still says Running.

```
$ kubectl get pod jenkins-0
```

NAME	READY	STATUS
jenkins-0	2/2	Running

```
$ kubectl logs jenkins-0 -c jenkins | grep -i space
```

```
java.io.IOException: No space left on device
```

BUILDS THAT NEVER CLEAN UP

Fills up over time

OLD BUILDS

Every run keeps its log and history forever.

BRANCHES AND PRS

Each one becomes a job with its own history.

ARTIFACTS

Packages archived on every build.

```
$ cd /var/jenkins_home
$ du -sh jobs/release-build/builds/*/archive
1.8G  jobs/release-build/builds/1/archive
1.8G  jobs/release-build/builds/2/archive

# build 3, Archive stage
java.io.IOException: No space left on device
```


AFTER THE DISK FILLS UP

FIND

Look inside the volume

```
$ df -h /var/jenkins_home
$ cd /var/jenkins_home
$ du -xsh * | sort -h
```

FIX

Free space or grow

```
$ rm troubleshooting-fill
```

Delete old builds, or expand the PVC

PREVENT

Fix the cause

Keep the last N builds per job

```
logRotator(numToKeepStr: '20')
```

Store artifacts in S3, not on the controller

Alert before the volume is full

THANK YOU.